

out of the darkness

The double-slit experiment proved that light travels as a wave, but early 20th-century experiments showed that light is made up of particles. In the face of these irreconcilable findings, a useful resort is to develop mental models to imagine the unimaginable. One of these models revolves around waves of probability, which we look at in the opening of this section. Probability led to the idea that electrons were both particle and wave, which then led to the conclusion that a quantum system does not exist in a definite state of either particle or wave until it is measured, which then formed the basis of the idea of uncertainty. Paradoxically, these findings about the apparently impossible nature of energy and matter led to quantum electrodynamics (QED), a theory that can account for all interactions between electrons and photons, including how the Sun shines, and what holds DNA together. It has also led to the development of lasers and microchips.



diffracted waves
Monochromatic
(single-wavelength)
light from a laser
spreads out in a
diffraction pattern as
it passes through an
aperture, showing
the wave nature
of light.